

1. Administrative & Study Identification

Full Study Title: A Phase 2, Multi-Center, Open-Label, Dual-Cohort Study to Evaluate the Efficacy and Safety of Lorlatinib (PF-06463922) Monotherapy in ALK Inhibitor-Treated Locally Advanced or Metastatic ALK-Positive Non-Small Cell Lung Cancer Patients in China **Short Title/Acronym:** A Study of Lorlatinib in ALK Inhibitor-Treated ALK-Positive NSCLC in China **Protocol Number:** B7461024 **NCT Number:** NCT03909971 **Posting ID:** 546115213648125496 **Sponsor/Company Name:** Pfizer **Investigational Product:** Lorlatinib (PF-06463922) | **Trade Name:** Lorbrena | **ATC Code:** L01ED05 **Phase of Development:** Phase II **Trial Status:** COMPLETED **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the objective response rate (ORR) by independent central radiology (ICR) according to Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1 in Cohort 1 (patients with previous crizotinib treatment). **Secondary Objectives:** To evaluate ORR in Cohort 2, Progression-Free Survival (PFS), Overall Survival (OS), Intracranial ORR, Time to Intracranial Progression, and safety/ tolerability (including treatment-related adverse events).

2.2 Background & Rationale To address the critical unmet medical need in Chinese patients with locally advanced or metastatic ALK-positive Non-Small Cell Lung Cancer (NSCLC) who have progressed on previous ALK tyrosine kinase inhibitors (TKIs). Lorlatinib, a brain-penetrant, third-generation ALK inhibitor, demonstrated robust global activity, and this study evaluates its targeted efficacy and safety specifically within the Chinese patient population, focusing heavily on durable control of brain metastases.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm (Dual-cohort; historically controlled/non-comparative) **Blinding:** Open-label **Randomization:** N/A (Assigned to cohorts based on prior therapy)

3.2 Planned Enrollment & Duration **Target N\$:** 109 enrolled patients (Cohort 1: \$n=67\$; Cohort 2: \$n=42\$) **Number of Sites:** 21 locations across China **Estimated Study Start:** 28-Apr-2019 **Study Status:** Completed

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Evidence of histologically or cytologically confirmed diagnosis of locally advanced or metastatic ALK-positive NSCLC where ALK status has been previously established by validated assays (e.g., Ventana ALK (D5F3) CDx, Vysis ALK Break Apart FISH, or AmoyDx EML4 ALK Fusion Gene Detection Kit). 2. Cohort 1: Disease progression after crizotinib as the only ALK inhibitor; Cohort 2: Disease progression after one ALK inhibitor other than crizotinib, with or without prior crizotinib. 3. All subjects must have at least 1 measurable extracranial target lesion according to RECIST v1.1 that has not been previously irradiated.

4.2 Exclusion Criteria 1. Prior therapy with an antibody or drug specifically targeting T-cell co-stimulation or immune checkpoint pathways (e.g., anti-PD-1, anti-PD-L1, anti-CTLA-4). 2. Major surgery within 4 weeks prior to enrollment, or radiation therapy within 2 weeks prior to enrollment. 3. Known EGFR activating mutations; prior therapy with EGFR TKIs; or more than 1 prior chemotherapy regimen in the advanced/metastatic setting.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Lorlatinib (Lorbrena) 100 mg once daily. **Route of Administration:** Oral. **Duration of Treatment:** Continuous 21-day cycles until disease progression, unacceptable toxicity, or withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Routine palliative and supportive care, stable or decreasing dose of ≤ 10 mg QD prednisone or equivalent for asymptomatic CNS metastases. **Prohibited:** Known strong CYP3A inducers (e.g., carbamazepine, phenytoin, rifampin, St. John's Wort) and P-glycoprotein (P-gp) substrates with a narrow therapeutic index.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Assessment (CT/MRI), Labs
Baseline	Day 0	First Dose, Vital Signs, Safety Labs
Treatment	Day 1 of 21-Day Cycles	Safety Labs, Efficacy Assessment, Education, AE monitoring (TEAEs)
End of Study	Up to 36+ Months	Final Vital Signs, Exit Interview, IP Accountability, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percentage of Participants With Objective Response (Cohort 1): Objective response rate (ORR), defined as the percentage of participants with a best overall confirmed response of complete response (CR) or partial response (PR) according to Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1. **Measurement Method:** Independent Central Radiology (ICR) for disease progression assessment.

7.2 Secondary & Exploratory Endpoints 1. Percentage of Participants With Objective Response (Cohort 2) by ICR 2. Progression-Free Survival (PFS) Based on ICR Assessment 3. Progression-Free Survival (PFS) Based on Investigator Assessment 4. Overall Survival (OS), Intracranial Objective Response Rate, and Time to Intracranial Progression

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: AEs graded by NCI CTCAE v4.03, focusing on Treatment-Emergent Adverse Events (TEAEs) such as hypercholesterolemia and hypertriglyceridemia. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeframe; specifically monitoring for Grade 3/4 AEs and those leading to treatment discontinuation. **Data Monitoring Committee (DMC):** Internal safety monitoring and independent central review mechanisms managed by Pfizer.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) / Safety Set (all patients receiving ≥ 1 dose of lorlatinib). **Sample Size Justification:** Target of 109 subjects appropriately powered to evaluate precise ORR and PFS differences relative to historical controls. **Handling of Missing Data:** Standard survival analysis methods including Kaplan-Meier estimates for PFS and OS, with censoring based on the last adequate tumor assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring for safety and protocol compliance. **Audit Trail:** Data verification, eCRF locking, and rigorous Independent Central Radiology (ICR) review of all imaging scans.

11. Study Identifiers & Governance

Primary ID: B7461024 **Registry Number:** NCT03909971 **Posting ID:** 546115213648125496 **Sponsor/Lead Organization:** Pfizer **Study Title:** A Study of Lorlatinib in ALK Inhibitor-Treated ALK-Positive NSCLC in China **Official Regulatory Title:** A Phase 2, Multi-Center, Open-Label, Dual-Cohort Study to Evaluate the Efficacy and Safety of Lorlatinib (PF-06463922) Monotherapy in ALK Inhibitor-Treated Locally Advanced or Metastatic ALK-Positive Non-Small Cell Lung Cancer Patients in China

12. Clinical Framework & Disease Characteristics

Primary Condition / Disease Area: Carcinoma, Non-Small-Cell Lung **Preferred UMLS Name:** ALK-Positive Non-Small Cell Lung Carcinoma **Preferred Name:** Carcinoma **Disease Definition:** A malignant tumor arising from epithelial cells. Carcinomas that arise from glandular epithelium are called adenocarcinomas, those that arise from squamous epithelium are called squamous cell carcinomas, and those that arise from transitional epithelium are called transitional cell carcinomas. **Diagnosis Threshold:** ALK-positive confirmed by validated assay (e.g., Ventana D5F3 CDx, Vysis FISH, or AmoyDx) **Medical Methodology:** ALK Tyrosine Kinase Inhibitor (TKI) Therapy **Optimization Target:** Target lesion reduction per RECIST v1.1 and durable intracranial control

Disease Classification Codes: * **MeSH Code:** D002277 * **HPO Code:** HP:0030731 * **SNOMED ID:** 1187425009 * **SNOMED Hierarchy:** 1187225007 | Malignant epithelial neoplasm (morphologic abnormality); 118285006 | Epithelial neoplasm (morphologic abnormality); 108369006 | Neoplasm (morphologic abnormality); 1240414004 | Malignant neoplasm (morphologic abnormality)

13. Study Procedures & Methodology

Management Pathway: Lorlatinib Monotherapy continuous until disease progression or unacceptable toxicity **Education Protocol (HCP):** Protocol training and RECIST v1.1 tumor assessment standardization **Education Protocol (Patient):** Oral dosing adherence instructions and AE self-reporting (e.g., lipid profile changes) **Quality Audit Mechanism:** Independent Central Radiology (ICR) review of imaging and routine safety data auditing

14. Observation & Follow-Up Schedule

Total Duration: Continuous until progression, up to ~3.5 years long-term survival follow-up **Onsite Visit Cadence:** Every 21 days (Day 1 of each treatment cycle) **Remote Monitoring Frequency:**

Survival follow-up ongoing every 12 weeks post-progression
Checklist Review Interval: End of each 21-day cycle

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				